

NAVARI YASHWANTH REDDY

AI/ML Engineer | Data Science Undergraduate | Computer Vision | Generative AI

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Science undergraduate and AI/ML Engineer with hands-on experience designing and developing intelligent applications across **Machine Learning, Deep Learning, Computer Vision, Generative AI, NLP, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, and MLOps**. Experienced in building end-to-end AI pipelines involving **data preprocessing, exploratory data analysis, feature engineering, model training, validation, evaluation, inference optimization, semantic retrieval, LLM orchestration, and production deployment**. Proficient in **Python, TensorFlow, PyTorch, Keras, OpenCV, Scikit-learn, LangChain, Google Gemini, FAISS, FastAPI, Docker, PostgreSQL, and Git**. Developed real-time facial emotion recognition, YOLO-based object detection, multimodal agricultural AI, unsupervised NLP clustering, and enterprise-oriented multi-agent customer-support systems.

EXPERIENCE

AI Web Development Intern – InAmigos Foundation

2025

Remote

- Developed responsive web applications using **React.js, HTML5, CSS3, and JavaScript**, integrating AI-powered functionality into web-based workflows.
- Built and integrated backend **REST APIs** using **Node.js and PostgreSQL** for volunteer management, donation processing, and application data workflows.
- Implemented AI and machine-learning functionality using **Python**, contributing to intelligent features and automation within NGO-oriented applications.
- Developed volunteer management workflows, donation modules, responsive interfaces, and database-backed application components.
- Worked in collaborative software-development environments using **Git, GitHub, Docker, and Agile development practices**.

AI Engineer / Full-Stack AI Developer – Innoexis

2026

AI Internship Project

- Developed **SupportSphere AI**, an enterprise-oriented AI customer-support platform combining **Generative AI, multi-agent orchestration, RAG, semantic routing, knowledge retrieval, and real-time conversational AI**.
- Designed a supervisor-based architecture coordinating **six specialized AI agents** for technical support, billing, orders, FAQs, complaints, and escalation workflows.
- Implemented a semantic knowledge-retrieval pipeline using **BGE embeddings, FAISS vector search, document chunking, Top-K retrieval, similarity search, and context injection**.
- Developed asynchronous **FastAPI** services and **WebSocket** communication for real-time bidirectional conversational AI and streaming responses.
- Integrated **Google Gemini** for LLM inference and implemented authentication, persistence, analytics, Docker-based deployment, and **OpenTelemetry** observability.

SELECTED AI/ML PROJECTS

SupportSphere AI – Autonomous Multi-Agent Customer Support *Python / FastAPI / LangChain / Gemini / FAISS / RAG*

- Architected an AI customer-support platform combining **multi-agent LLM orchestration, semantic intent routing, RAG retrieval, response synthesis, confidence-based escalation, and human handoff workflows**.
- Designed a supervisor architecture coordinating **six specialized AI agents** for technical support, billing, orders, FAQ, complaints, and escalation.
- Built a knowledge-base retrieval pipeline using **document chunking, 384-dimensional BGE embeddings, FAISS vector indexing, Top-K similarity search, and context injection** to provide grounded LLM responses.
- Implemented asynchronous **FastAPI** services with WebSockets for real-time conversational streaming and bidirectional client-server communication.
- Integrated **Google Gemini** with LangChain-based orchestration and developed structured AI workflows for query classification, retrieval, response generation, and escalation.
- Implemented JWT authentication, SQLAlchemy persistence, PostgreSQL-backed application data, Docker/Docker Compose deployment, analytics, and **OpenTelemetry** tracing for production-oriented observability.

AgriSathi – AI Smart Farmer Assistant

React / TypeScript / Gemini / Multimodal AI / Computer Vision

- Built an AI-powered agricultural assistant combining **multimodal AI, computer vision, conversational AI, crop disease analysis, farm management, market intelligence, and agricultural decision support**.
- Integrated **Google Gemini** for image-based crop disease analysis and natural-language agricultural assistance.
- Developed AI workflows supporting crop disease diagnosis, treatment recommendations, farming questions, soil-health guidance, crop lifecycle management, and agricultural decision support.

- Designed an interactive application with AI dashboards, analytics, persistent application state, voice input, and responsive user interfaces.
- Implemented a production deployment workflow using **GitHub Actions** and modern frontend tooling.

DeepFER-Live – Real-Time Facial Emotion Recognition *Python / TensorFlow / Keras / CNN / OpenCV / FER-2013*

- Designed and trained a custom **Convolutional Neural Network (CNN)** using the **FER-2013 dataset** for seven-class real-time facial emotion recognition.
- Implemented an end-to-end computer vision pipeline covering **face detection, grayscale conversion, histogram equalization, 48×48 image preprocessing, CNN inference, and Softmax probability prediction.**
- Applied **data augmentation, convolutional layers, MaxPooling, dropout, and early stopping** to improve model generalization and training stability.
- Implemented temporal smoothing across consecutive frames to reduce prediction flicker and improve the stability of real-time emotion classification.
- Integrated OpenCV webcam processing with optimized inference targeting real-time CPU execution and frame processing below **50 ms.**
- Developed a complete training-to-inference workflow covering dataset preparation, model training, evaluation, saved-model inference, and live webcam prediction.

ML Object Detection System – Real-Time Object Detection *Python / YOLOv8 / PyTorch / OpenCV / Roboflow*

- Developed a real-time deep-learning object detection system using **YOLOv8 Nano** for simultaneous object localization and classification.
- Implemented an end-to-end training workflow involving **dataset preparation, configuration, model fine-tuning, weight generation, validation, and inference.**
- Integrated **Roboflow** for dataset management and automated dataset retrieval within the computer vision workflow.
- Used **PyTorch-based model weights and Ultralytics tooling** for model training and inference.
- Built OpenCV-based webcam inference for real-time object detection, bounding-box visualization, and prediction display.

Netflix Movies & TV Shows Clustering – Unsupervised ML *Python / Scikit-learn / NLP / TF-IDF / K-Means / PCA / SVD*

- Built an end-to-end **unsupervised machine learning pipeline** analyzing **7,787 Netflix titles** using textual metadata and content attributes.
- Performed **data cleaning, exploratory data analysis, missing-value analysis, categorical processing, text preprocessing, feature engineering, and dataset preparation.**
- Combined genres, descriptions, cast, directors, countries, and content type into engineered textual representations for machine-learning clustering.
- Applied **TF-IDF with 5,000 features**, Truncated SVD, L2 normalization, and K-Means clustering to transform high-dimensional text data into machine-learning-ready representations.
- Selected **K=6** using Elbow Method and Silhouette Analysis and evaluated cluster quality using Silhouette, Davies-Bouldin, and Calinski-Harabasz metrics.
- Performed cluster profiling and visualization to identify content patterns relevant to recommendation, content discovery, and catalog analysis.

Age & Gender Detection AI – Computer Vision *Python / OpenCV / DNN / Gradio / Pyodide / WebAssembly*

- Developed a real-time computer vision application for **face detection, age-group classification, and gender classification** from live webcam input.
- Integrated OpenCV DNN models for face detection and demographic prediction across continuously captured video frames.
- Built an interactive **Gradio** interface for real-time model interaction and prediction visualization.
- Explored browser-based execution using **Pyodide and WebAssembly**, enabling local inference without requiring a traditional remote backend.
- Designed the application around privacy-preserving local processing and real-time computer vision inference.

ADDITIONAL AI & DATA PROJECTS

- **NeuroVision ML Platform** – Designed and trained custom object-detection models using dataset creation, annotation, data augmentation, YOLO architecture fine-tuning, and model evaluation.
- **Heart Disease Risk Prediction System** – Developed a supervised machine-learning solution for predicting heart disease from patient medical parameters.
- **Smart Campus Network** – Built an intelligent campus network monitoring platform with automated device access control, authentication-event processing, PostgreSQL data management, and real-time analytics.
- **Cricbuzz LiveStats** – Developed a real-time cricket analytics dashboard using Python, PostgreSQL, Streamlit, SQL, and the Cricbuzz RapidAPI.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, SQL

Machine Learning: Machine Learning, Supervised Learning, Unsupervised Learning, Classification, Clustering, Regression,

Feature Engineering, Model Evaluation, Hyperparameter Tuning, Exploratory Data Analysis

Deep Learning: Neural Networks, CNNs, Deep Learning, Transfer Learning, Model Training, Data Augmentation, Dropout, MaxPooling, Early Stopping, Softmax, Model Inference

Generative AI: Generative AI, Large Language Models, LLMs, Google Gemini, LangChain, Multi-Agent Systems, Prompt Engineering, RAG, Retrieval-Augmented Generation, Semantic Search, Vector Embeddings, FAISS

Computer Vision: OpenCV, YOLOv8, PyTorch, TensorFlow, Keras, Image Classification, Object Detection, Face Detection, Facial Emotion Recognition, Real-Time Video Processing

Natural Language Processing: NLP, Text Preprocessing, TF-IDF, Text Embeddings, Semantic Similarity, SVD, PCA, Dimensionality Reduction, NLP Feature Engineering

Data Science: Pandas, NumPy, Scikit-learn, Matplotlib, Data Cleaning, Data Preprocessing, EDA, Data Visualization, Statistical Analysis

AI Engineering: AI Agents, Agent Orchestration, Knowledge Retrieval, Vector Databases, Semantic Routing, Context Injection, Confidence-Based Escalation, AI Workflows

MLOps & Deployment: Docker, Docker Compose, GitHub Actions, OpenTelemetry, REST APIs, WebSockets, FastAPI, Flask, Streamlit, Model Deployment, Application Monitoring

Backend & Databases: FastAPI, Flask, Node.js, Express.js, PostgreSQL, MySQL, SQLite, SQLAlchemy, REST APIs

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Power BI, Vite

EDUCATION

Woxsen University

Bachelor of Technology (B.Tech.) in Data Science

2023 – 2027

Hyderabad, Telangana

Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Computer Networks, Cloud Computing, Operating Systems, Software Engineering

CERTIFICATIONS

- **Java Programming** – Duke University, Coursera
- **Object-Oriented Programming** – University of Alberta, Coursera
- **Data Structures in Python** – Coursera

ACHIEVEMENTS

- **Government of India Design Patent Holder** – Patent No. 470097-001 for an innovative IoT Connectivity Device.
- Built and deployed **20+ software projects** spanning Artificial Intelligence, Machine Learning, Computer Vision, Data Science, Healthcare, Networking, and Full-Stack Engineering.
- Developed publicly accessible AI/ML applications demonstrating **model development, real-time inference, computer vision, Generative AI, semantic retrieval, and production deployment.**
- Maintains public GitHub repositories covering **Generative AI, RAG, multi-agent systems, computer vision, deep learning, NLP, and classical machine learning.**

PROFILES

Portfolio: navariyashwanthreddy.vercel.app

GitHub: github.com/YashwanthNavari

LinkedIn: linkedin.com/in/navari-yashwanth-reddy-4a7065357

Selected Repositories:

[SupportSphere AI](#) | [AgriSathi](#) | [DeepFER-Live](#) | [Object Detection](#)

[Netflix Clustering](#) | [Age & Gender Detection](#) | [NeuroVision](#)